

Distinguishing Medical Diversity from Byzantine Attacks: Client-Adaptive Anomaly Detection for Healthcare Federated Learning

Research in Progress

Introduction

Federated learning (FL) enables healthcare institutions to collaboratively train machine learning models without sharing sensitive patient data, addressing critical privacy regulations like HIPAA and GDPR (McMahan et al. 2017). Despite its promise, FL’s distributed nature exposes it to Byzantine failures —where compromised or faulty participants submit corrupted gradient updates that can catastrophically degrade model performance (Toa et al, 2023). In healthcare contexts, where model errors directly impact patient outcomes, even a single Byzantine participant in a multi-hospital network could cause significant increases in misdiagnosis rates (Bagdasaryan et al. 2020).

The Byzantine robustness challenge is particularly acute in healthcare due to data heterogeneity across institutions. Pediatric hospitals generate fundamentally different gradient updates than geriatric centers, creating natural statistical diversity that existing defenses struggle to distinguish from adversarial behavior. Current Byzantine-robust aggregation methods apply uniform filtering rules that treat all deviations as suspicious. Recent experimental studies demonstrate severe performance degradation in heterogeneous settings, with some methods achieving less than 10% accuracy even without attacks (Li, Ngai, and Voigt 2024), effectively excluding valuable

contributions from specialized medical centers and undermining the collaborative benefits of federated learning.

We propose CAAC-FL (Client-Adaptive Anomaly-Aware Clipping), a novel defense mechanism that fundamentally reimagines Byzantine detection through client-specific behavioral monitoring. Unlike existing methods that apply global server-side thresholds, CAAC-FL maintains node-level behavioral profiles and adapts clipping thresholds based on each client’s historical consistency patterns rather than population-wide statistics. Our client-adaptive strategy moves beyond one-size-fits-all defenses, providing the ability to separate authentic institutional variability from adversarial manipulation—an unresolved challenge highlighted in recent studies (Werner et al. 2023).

Related Work and Limitations

Existing Byzantine-robust federated learning methods fall into three categories, each with limitations in heterogeneous healthcare settings. Statistical filtering approaches like Trimmed Mean (Yin et al. 2018) remove outliers coordinate-wise but fail when Byzantine clients craft updates appearing normal in most dimensions while being malicious in critical ones. Geometric methods such as Krum (Blanchard et al. 2017) select “central” updates based on distance metrics, incorrectly filtering legitimate but diverse contributions from specialized institutions like pediatric versus geriatric centers. Recent adaptive methods, including ARC (Baruch, Baruch, and Goldberg 2019) and RFA (Pillutla, Kakade, and Harchaoui 2021), automatically determine clipping thresholds but still apply global rules that cannot accommodate persistent institutional differences.

FLTrust (Cao et al. 2021) introduced trust bootstrapping using a server-side “root dataset,” but this violates federated learning’s core privacy principle by requiring centralized data. Recent work on Byzantine-resilient multi-task learning (Le and Moothedath 2025) explores client-specific model layers but lacks temporal behavioral tracking needed to distinguish sudden attacks from gradual institutional evolution.

The fundamental limitation is treating all clients as statistically identical. Li et al. (2024) demonstrated that existing defenses achieve less than 10% accuracy in non-IID settings even without attacks, as global thresholds cannot distinguish legitimate diversity from Byzantine behavior. While recent theory (Werner et al. 2023) proves that client-specific adaptation enhances robustness, practical implementations of this theory remain nascent. This gap between theoretical proofs and working systems motivates our client-specific behavioral profiling approach, as shown in Table 1’s comparison with recent methods.

Method	Description	Magnitude Defense	Directional Defense	Client-Specific	Temporal	Adaptive
Median/Trimmed Mean (Yin et al. 2018)	Coordinate-wise statistical filtering	✓ (coordinate)	✗	✗	✗	✗
Krum (Blanchard et al. 2017)	Selects gradient with minimum sum of distances	✓ (L2 distance)	✗	✗	✗	✗
FLTrust (Cao et al. 2021)	Trust scores based on server root dataset	✓ (norm)	✓ (cosine)	✗	✗	✗
Clipped Clustering (Li, Ngai, and Voigt 2024)	Clips then clusters based on similarity	✓ (median norm)	✓ (cosine clustering)	✗	✗	✓ (global)
LASA (Xu, Hu, et al. 2024)	Layer-wise sparsification and filtering	✓ (layer norm)	✓ (layer direction)	✓ (sparsification)	✗	✓ (layer)
BR-MTRL (Le and Moothedath 2025)	Multi-task with client-specific layers	✓ (task-specific)	✗	✓ (final layer)	✗	✗

Method	Description	Magnitude Defense	Directional Defense	Client-Specific	Temporal	Adaptive
CAAC-FL (ours)	Client-adaptive temporal behavioral profiling	✓ (norm + EWMA)	✓ (cosine + median)	✓ (node-level)	✓ (EWMA)	✓ (exponential)

Table 1: Comparison with recent contributions

CAAC-FL uniquely combines comprehensive defense capabilities. The magnitude anomaly detection catches scaling attacks like ALIE (‘A Little is Enough’ (Baruch, Baruch, and Goldberg 2019)), while directional anomaly detection identifies steering attempts such as inner product manipulation (Cong, Koyejo, and Gupta 2019; Fang et al. 2020). The composite score makes it harder for attackers since they must appear normal across all three metrics simultaneously, not just one.

Proposed Approach: CAAC-FL

We consider N healthcare institutions with private datasets $\mathcal{D}_i$. At round t , clients compute gradient updates $g_i^{(t)}$, with fraction $\alpha < 0.5$ potentially Byzantine. Byzantine behaviors include both intentional attacks (gradient manipulation, data poisoning) and unintentional faults (hardware degradation, corrupted data)—both indistinguishable at the aggregation level.

CAAC-FL’s key innovation lies in the concept of maintaining temporal behavioral profiles for each participating node. For client i at round t , we propose tracking a profile $H_i^{(t)}$ containing: (1) historical gradient norms $\{\|g_i^{(k)}\|_2\}_{k=t-W}^{t-1}$ over a sliding window W , (2) directional consistency measurements $\cos(g_i^{(k)}, g_i^{(j)})$ for recent update pairs, and (3) exponentially weighted moving average (EWMA) statistics where $\mu_i^{(t)} = \alpha \cdot \|g_i^{(t)}\|_2 + (1 - \alpha) \cdot \mu_i^{(t-1)}$ for the mean and $\sigma_i^{(t)} =$

$\sqrt{\alpha \cdot \left(\|g_i^{(t)}\|_2 - \mu_i^{(t-1)} \right)^2 + (1 - \alpha) \cdot \left(\sigma_i^{(t-1)} \right)^2}$ for the standard deviation, with decay parameter α controlling the emphasis on recent versus historical behavior. This approach will capture each client's unique "gradient signature" reflecting their data distribution and clinical focus, potentially enabling distinction between consistent institutional patterns and anomalous deviations.

Preventing early profile contamination requires a bootstrap phase with conservative uniform clipping followed by gradual transition to client-specific thresholds. Key parameters including bootstrap duration, stability metrics, and transition criteria require empirical tuning.

We propose computing anomaly scores across three dimensions to capture different types of Byzantine behavior. The magnitude anomaly would measure deviation from the EWMA-based historical norms:

$A_{mag}^{(i,t)} = \frac{|\|g_i^{(t)}\|_2 - \mu_i^{(t-1)}|}{\sigma_i^{(t-1)} + \epsilon}$. The directional anomaly would capture sudden

directional shifts: $A_{dir}^{(i,t)} = 1 - \frac{1}{W} \sum_{k=t-W}^{t-1} \cos(g_i^{(t)}, g_i^{(k)})$, where k indexes past rounds within the

window. The temporal consistency score would detect distribution changes: $A_{temp}^{(i,t)} = \frac{|\sigma_i^{(t)} - \sigma_i^{(t-W)}|}{\sigma_i^{(t-W)} + \epsilon}$,

measuring variance drift over time. These three dimensions would be combined into a composite

anomaly score $A_i^{(t)} = \sqrt{\sum_j \lambda_j \left(A_j^{(i,t)} \right)^2}$ where $j \in \{mag, dir, temp\}$ and the weights λ_j are key parameters to optimize experimentally.

Client-specific clipping thresholds would adapt based on behavioral consistency. We propose a threshold function of the form $\tau_i^{(t)} = \mu_{global}^{(t)} \cdot f(A_i^{(t)}, R_i^{(t)})$ where $\mu_{global}^{(t)}$ is the global median

norm, $A_i^{(t)}$ is the anomaly score, and $R_i^{(t)}$ represents historical reliability computed as an exponentially weighted moving average: $R_i^{(t)} = \gamma \cdot \mathbb{1}(A_i^{(t)} < \tau_{anomaly}) + (1 - \gamma) \cdot R_i^{(t-1)}$, where $\mathbb{1}(\cdot)$ equals 1 if the condition is true and 0 otherwise, $\tau_{anomaly}$ is an anomaly threshold, and $\gamma \in (0,1)$ is the smoothing parameter. The function f would decrease with higher anomaly scores (suggesting stricter clipping for suspicious clients) and increase with higher reliability scores (allowing more flexibility for consistently trustworthy clients). The specific form of f (e.g., exponential, sigmoid) and its parameters require empirical validation.

The integration of new institutions presents a key challenge requiring careful balance between security and inclusivity. While we envision a graduated trust establishment process—beginning with conservative uniform clipping before transitioning to client-specific profiling—the precise bootstrap duration and transition metrics require further investigation.

Research Hypotheses

- *H1 (Heterogeneity Preservation)*: Client-specific behavioral profiles will significantly reduce false positive rates compared to global threshold methods when legitimate clients have heterogeneous data distributions, while maintaining comparable or better Byzantine detection rates.
- *H2 (Multi-Dimensional Defense)*: Combining magnitude, directional, and temporal anomaly metrics will provide more robust Byzantine detection than single-metric approaches,

particularly against adaptive adversaries that attempt to evade detection by mimicking normal behavior in individual dimensions.

- *H3 (Temporal Discrimination)*: The window-based profiling approach will successfully distinguish between abrupt Byzantine attacks and gradual legitimate institutional changes, addressing a key limitation of existing methods that either over-react to legitimate evolution or under-react to slow poisoning attacks.

These hypotheses directly address the fundamental tension in heterogeneous federated learning: maintaining security without excluding legitimate diversity. Our experimental plan is designed to test each hypothesis across diverse healthcare scenarios rigorously.

Experimental Plan

We will evaluate CAAC-FL on three healthcare datasets: MIMIC-III (Johnson et al. 2016) for ICU mortality prediction ($n=49,785$), ChestX-ray8 (Wang et al. 2017) for multi-label disease classification (108,948 images), and ISIC 2019 (Codella et al. 2018) for melanoma detection ($n=2,750$). We simulate federated settings with 20 clients, introducing heterogeneity through Dirichlet allocation ($\alpha=0.5$) for label skew, power law dataset sizes, and domain-specific augmentations (Hsu et al. 2019).

We implement untargeted attacks (random noise, sign flipping), targeted attacks (ALIE (Baruch, Baruch, and Goldberg 2019), Inner Product Manipulation (Fang et al. 2020)), and adaptive attacks, including slow-drift poisoning and profile-aware strategies. Byzantine fractions range from 20% to 40%, testing both independent and coordinated strategies.

Primary metrics include model accuracy, false positive rate (testing H1), attack impact (testing H2), and detection latency (testing H3). We compare against FedAvg (McMahan et al. 2017), Krum (Blanchard et al. 2017), Trimmed Mean (Yin et al. 2018), ARC (Baruch, Baruch, and Goldberg 2019), FLTrust (Cao et al. 2021), and LASA (Xu, Hu, et al. 2024)). The design targets our hypotheses through heterogeneous distributions, diverse attack strategies, and temporal patterns, with benefits expected under high heterogeneity and 20-30% Byzantine fractions.

Discussion and Future Work

This work-in-progress presents CAAC-FL, addressing the fundamental challenge of distinguishing legitimate heterogeneity from adversarial behavior in federated learning. By maintaining client-specific behavioral profiles rather than applying global thresholds, CAAC-FL enables secure collaboration while preserving institutional diversity—essential for healthcare where uniform security measures either compromise safety or exclude valuable participants.

The approach makes four key contributions: (1) a client-adaptive framework maintaining node-specific behavioral profiles, among the first practical implementations of recent theoretical work (Werner et al. 2023); (2) multi-dimensional anomaly detection combining magnitude, direction, and temporal metrics; (3) adaptive thresholding balancing responsiveness with stability; and (4) a comprehensive experimental design for healthcare-specific validation. Our three hypotheses directly address critical gaps: reducing false positives through client-specific profiling, enhancing detection via multi-dimensional scoring, and distinguishing sudden attacks from gradual legitimate changes.

Key limitations include vulnerability to slow-drift attacks, potential manipulation by colluding Byzantine clients, scalability challenges for thousands of participants, and privacy risks from detailed behavioral profiles. Future work should address formal convergence guarantees, differential privacy integration, hierarchical aggregation for large-scale deployments, and cross-domain validation beyond healthcare. Despite these challenges, CAAC-FL represents a paradigm shift toward context-aware Byzantine defense, with implications for any heterogeneous federated learning deployment where both security and inclusivity are paramount.

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